Hi 

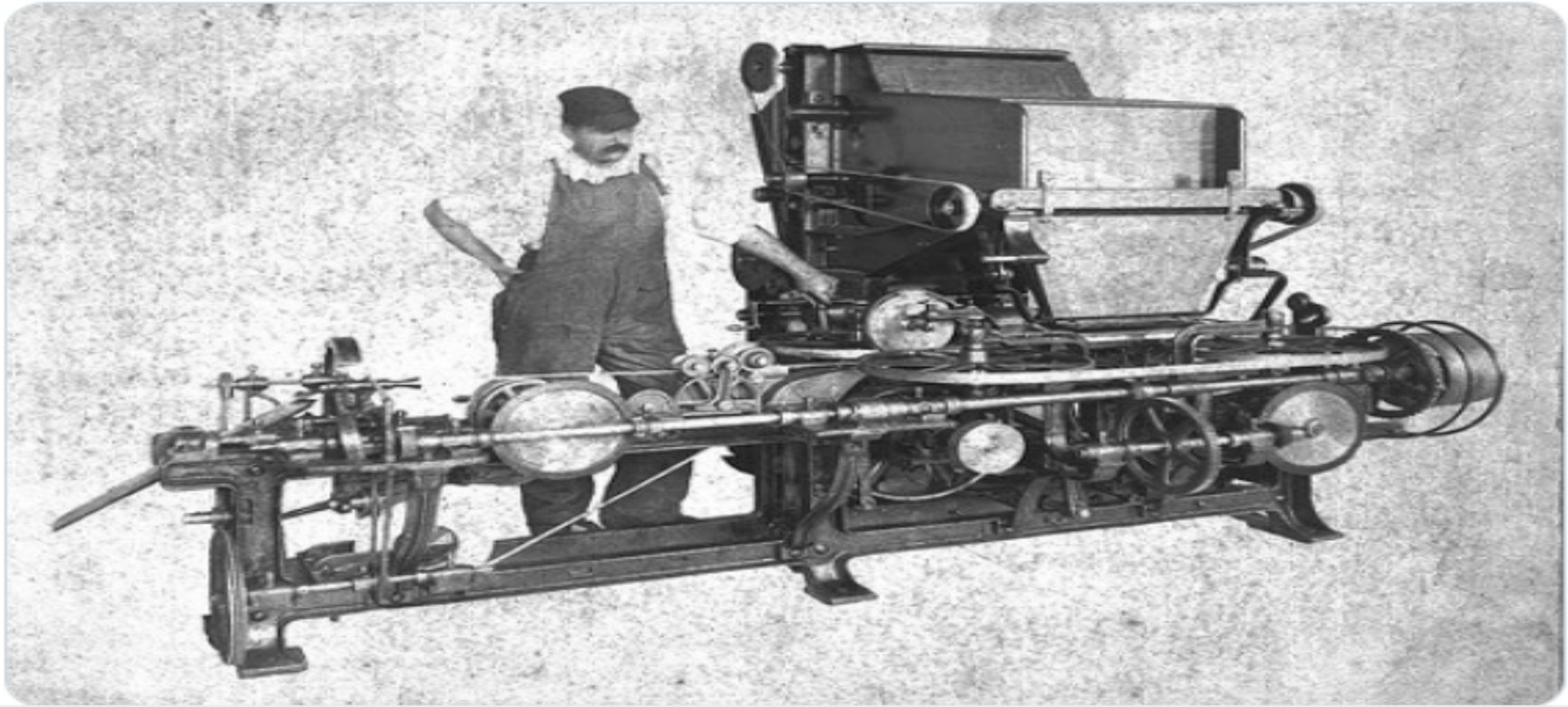
A SOTA OVERVIEW OF EFFICIENT AI

What's working, what's hype, and what matters for Africa

Steven Kolawole

IndabaX Nigeria 2026 · University of Ibadan

A rare archival photo of a machine learning engineer fine tuning an early version of XGBoost in the 1880s.



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Abeokuta Grammar School | CS @ Federal University of Agriculture, Abeokuta



A BIT ABOUT ME

00 **Abeokuta Grammar School | CS @ Federal University of Agriculture, Abeokuta**

01 **Doctoral Researcher, CMU SCS — Language Technologies Institute**

02 **Research: making large models cheaper and faster to serve**

03 **AWS Bedrock — Inference Optimization Science Team**

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03 **AWS Bedrock — Inference Optimization Science Team**

04 **Founding Organizer, ML Collective Africa**

Most of what we hear about AI is solving someone else's problem.

THREE BUCKETS

1 **Imported debates**

2 **Misdirected ambitions**

3 **Technical mythology**

TWELVE CLAIMS

What's hype, what's real

BUCKET 1

Imported debates

Someone else's questions

1. "AGI is 2 to 5 years away."
2. "AI is going to take all the jobs."
3. "AI is an existential threat we should be organizing around."
4. "The frontier labs will solve this for everyone."

BUCKET 2

Misdirected ambitions

Wrong layer of the stack

5. "Africa needs to train its own foundation models to be sovereign."
6. "We need African GPUs and African compute."
7. "If we leverage open-source, we can catch up to frontier labs."
8. "If we just had bigger models, we'd solve African languages."

BUCKET 3

Technical mythology

Where decisions compound

9. "Small language models trained from scratch are the answer for African use cases."
10. "Fine-tuning is the move."
11. "Edge deployment is the future for Africa."
12. "Just call the OpenAI API. We'll optimize later."

Fast	through Imported
Slower	through Ambitions
Slowest	through Mythology

BUCKET 1

Imported debates

Someone else's questions distorting our priorities.

"AGI is 2 to 5 years away."

WHY PEOPLE BELIEVE IT

Frontier lab CEOs keep saying it. The capability jumps from GPT-4 to today feel real. Timelines have collapsed before.

WHAT'S ACTUALLY TRUE

Nobody knows.
There’s enormous gap between "AGI exists in a lab" and "AGI changes what you ship in Lagos next quarter".

IMPLICATION

Don't let a debate you can't resolve set your 18-month roadmap.

"AI is going to take all the jobs."

WHY PEOPLE BELIEVE IT

The framing is everywhere. And some jobs really are changing.

WHAT'S ACTUALLY TRUE

Job displacement is real but slow, uneven, and heavily mediated by which jobs your economy actually has.
The Nigerian labor market is not the SF labor market. Predictions calibrated to one don't transfer to the other.

IMPLICATION

Worry about the specific jobs in your specific economy, not the global average.

"AI is an existential threat we should be organizing around."

WHY PEOPLE BELIEVE IT

Serious researchers say it. The argument is internally coherent. Hard to dismiss

WHAT'S ACTUALLY TRUE

I'm not telling you what to think about x-risk. I'll tell you it's louder in our timelines than it deserves to be given what we're trying to build. Engage with it intellectually if you want — don't let it set your priorities.

IMPLICATION

Reserve your scarce attention for problems you can actually move.

"The frontier labs will solve this for everyone."

WHY PEOPLE BELIEVE IT

They keep shipping. APIs keep getting better and cheaper. Why reinvent what OpenAI is doing?

WHAT'S ACTUALLY TRUE

Frontier labs are solving for frontier-lab customers. Their roadmap is not your roadmap. Their pricing is not your pricing. Their language coverage is not your language coverage. "Eventually" is doing all the work in that sentence.

IMPLICATION

Plan for the world where they don't, because that's the world you're already in.

None of these change what we should do this year.

African builders have a 12–24 month planning horizon. The AGI debate is a luxury good.

BUCKET 2

Misdirected ambitions

Real concerns but aimed at the wrong layer.

"Africa needs to train its own foundation models to be sovereign."

WHY PEOPLE BELIEVE IT

Sovereignty matters. Data colonialism is real. Depending on Anthropic and OpenAI feels precarious. Pretraining is the visible, prestigious thing.

WHAT'S ACTUALLY TRUE

Sovereignty doesn't live at the pretraining layer for almost anyone. It lives at the data, fine-tuning, evaluation, and inference layers. Pretraining from scratch costs more than most African AI ecosystems can afford — and produces a worse model than fine-tuning an open-weight one.

IMPLICATION

Sovereignty is a stack question, not a pretraining question. Own the layers you can actually own.

"We need African GPUs and African compute."

WHY PEOPLE BELIEVE IT

Real concern about compute dependency. Sanctions and export controls have shown other countries this can become a vulnerability.

WHAT'S ACTUALLY TRUE

Long-term, yes. Short-term, the binding constraint isn't compute access — it's inference literacy. Most African teams aren't using the GPUs they already have access to efficiently. Building an African GPU industry takes 15 years. Teaching your team to run vLLM properly takes a weekend.

IMPLICATION

Compute sovereignty matters. Inference literacy matters more, right now, this year.

"If we leverage open-source, we can catch up to frontier labs."

WHY PEOPLE BELIEVE IT

Open-weight models really have caught up on benchmarks. Llama, Mistral, Qwen, DeepSeek — the gap looks closed if you're reading the leaderboards.

WHAT'S ACTUALLY TRUE

Benchmarks measure what benchmarks measure. The gap between open-weight and frontier shows up in the things that aren't on leaderboards — long-context reliability, tool use, safety, multilingual edge cases, the parts of product quality that don't fit in a number. Catching up on MMLU is not catching up on shipping.

IMPLICATION

Open-source is enough for most products you'll build. It is not "catching up." Use it for what it is.

"If we just had bigger models, we'd solve African languages."

WHY PEOPLE BELIEVE IT

Scale solved a lot of things. It feels like it should solve this too.

WHAT'S ACTUALLY TRUE

Scale doesn't solve data scarcity, evaluation scarcity, or representation gaps. A 70B model trained on the same Yoruba data as a 7B model gets you marginally better Yoruba and a much larger bill. The bottleneck is corpus and benchmarks, not parameter count.

IMPLICATION

Build the data and the evals. The model will follow.

BUCKET 3

Technical mythology

Where decisions actually compound.

"Small models trained from scratch are the answer for African use cases."

WHY PEOPLE BELIEVE IT

Smaller models, lower compute, "appropriate for our constraints" — it fits the resource-constrained narrative cleanly.

WHAT'S ACTUALLY TRUE

Distilling or fine-tuning an open-weight 7B model dominates training a 1B from scratch on almost every axis — quality, time, cost, maintainability. The "trained from scratch" framing confuses origin with ownership. You can own a model you didn't pretrain.

IMPLICATION

Reach for distillation and fine-tuning before reaching for pretraining. The right "small" model is usually a compressed big one.

"Fine-tuning is the move."

WHY PEOPLE BELIEVE IT

It's the most visible knob. Every tutorial reaches for it. It feels like real ML work.

WHAT'S ACTUALLY TRUE

Most teams reaching for fine-tuning should be reaching for better prompting and retrieval first. The hierarchy is: prompt engineering → retrieval-augmented generation → fine-tuning, in that order of cost and complexity. Fine-tuning is rarely the cheapest path to the quality you want.

IMPLICATION

Exhaust prompting and retrieval before fine-tuning. Most "we need to fine-tune" conversations end before fine-tuning is justified.

"Edge deployment is the future for Africa."

WHY PEOPLE BELIEVE IT

Connectivity is uneven. Privacy matters. Demos look great. The narrative is appealing.

WHAT'S ACTUALLY TRUE

Almost every African AI deployment that actually ships is server-side. The constraint isn't device compute — it's memory bandwidth, latency tolerance, and serving economics. Edge demos rarely make it to production. Pretending otherwise pushes researchers toward tiny-model work that doesn't ship.

IMPLICATION

Start with server-side. Move to edge only when you have a specific reason. The reason is almost never "Africa."

"Just call the OpenAI API. We'll optimize later."

WHY PEOPLE BELIEVE IT

It's the fastest path to a working prototype. Why prematurely optimize?

WHAT'S ACTUALLY TRUE

The architectural decisions you make assuming an API are very hard to reverse. Your prompts, retry logic, latency budgets, cost model, privacy posture — all of it gets shaped by the assumption that inference is someone else's problem. "Later" usually means never. And when the price changes, or the model deprecates, you find out how much of your product was actually theirs.

IMPLICATION

Every project should have a "what happens when we self-host" document, even if you never act on it.

WHERE WE LANDED

The twelve, revisited

Bucket 1 — set aside. Buckets 2 and 3 — answered, at least with one perspective.

BUCKET 1

Set aside

Not our fight today

1. "AGI is 2 to 5 years away."
2. "AI is going to take all the jobs."
3. "AI is an existential threat we should be organizing around."
4. "The frontier labs will solve this for everyone."

BUCKET 2

Answered

Aim lower in the stack

5. Sovereignty lives at inference, not pretraining.
6. Inference literacy before African GPUs.
7. Open-source is enough — not "catching up."
8. Build data and evals. Model follows.

BUCKET 3

Answered

This is where you ship

9. Distill and fine-tune. Don't pretrain.
10. Prompting → retrieval → fine-tuning.
11. Server-side first. Edge is the exception.
12. Plan for the day you self-host.

WHAT TO DO MONDAY

Concrete moves, ranked by horizon

DO THIS WEEK

01 **Audit your serving stack.**

If you're not running vLLM or SGLang with continuous batching, that's the first 80% of gains, free.

02 **Stop reaching for fine-tuning.**

Re-run your last fine-tuning conversation as: did we exhaust prompting and retrieval first?

03 **Quantize.**

INT8 or INT4 on an open-weight model. Unsexy, almost always pays off.

REVISIT THIS QUARTER

04 **Write the self-host doc.**

Even if you never act on it. Forces you to surface the architectural lock-in you've accumulated.

05 **Invest in evals before models.**

Especially for African-language work. The bottleneck is measurement, not parameters.

06 **Pick one stack-ownership skill.**

vLLM internals, KV cache management, or quantization calibration. Go deep on one.

While that debate runs,
the people who own their inference stack
will be the ones still standing
when the dust settles —
here or anywhere else.

I came to CMU thinking the most important thing I could do was research frontier compression. Two years in, I think the most important thing I can do is help more African teams own their inference stack. Both might be right. The second one is more urgent.

Deep Learning Indaba 2026 @ PAU

August · MLC Africa Workshop

Whose Intelligence?

Building Agentic AI in Africa
without Frontier Model Dependency

If today's talk resonated, come build with us in August.

Thank you.

Questions?

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